

1. List the key differences between IPv4 and IPv6 addressing by creating a two-column table with at least 4 points for each.

IPv4	IPv6
Uses 32-bit addresses.	Uses 128-bit addresses.
Example: 192.168.1.10	Example: 2001:db8::1
Has about 4.3 billion addresses.	Has a very large number of addresses.
Written in decimal numbers .	Written in hexadecimal numbers .
Uses NAT commonly to save IP addresses.	Usually does not need NAT because of the large address space.

2. Given the following scenario: You have 4 devices (laptop, phone, smart TV, and printer) connected to your home WiFi. Assign a unique static IPv4 address to each device using the 192.168.10.0/24 network, and write down the IP, subnet mask, and default gateway for each device.

Network: **192.168.10.0/24**

Subnet Mask: **255.255.255.0**

Default Gateway: **192.168.10.1**

Device	IP Address	Subnet Mask	Default Gateway
 Laptop	192.168.10.2	255.255.255.0	192.168.10.1
 Phone	192.168.10.3	255.255.255.0	192.168.10.1
 Smart TV	192.168.10.4	255.255.255.0	192.168.10.1
 Printer	192.168.10.5	255.255.255.0	192.168.10.1

3. Calculate the number of valid subnets and hosts per subnet if you use a /27 CIDR subnet mask on a 192.168.1.0 network. List the first three valid subnet address ranges (network address to broadcast address) and the usable host IPs in each.

- Network = **192.168.1.0**
- CIDR = **/27**

Step 1: Subnet Mask

/27 = 255.255.255.224

Step 2: Number of Subnets

Borrowed bits = **3**

$2^3 = 8$ Subnets

Step 3: Hosts per Subnet

Host bits = **5**

$2^5 = 32$ IP addresses

$32 - 2 = 30$ usable hosts

First Three Subnets

Subnet	Network Address	Broadcast Address	Usable Host Range
1	192.168.1.0	192.168.1.31	192.168.1.1 – 192.168.1.30
2	192.168.1.32	192.168.1.63	192.168.1.33 – 192.168.1.62
3	192.168.1.64	192.168.1.95	192.168.1.65 – 192.168.1.94

Final Answer

- **Subnet Mask:** 255.255.255.224
- **Valid Subnets:** 8
- **Usable Hosts per Subnet:** 30

4.You are given the IP address 10.0.5.17 with subnet mask 255.255.255.0. Determine the network address, broadcast address, and the range of valid host addresses for this subnet.

Hint: Convert the subnet mask to CIDR notation and use bitwise AND for network address calculation.

IP Address = **10.0.5.17**

Subnet Mask = **255.255.255.0**

Step 1: CIDR Notation

255.255.255.0 = /24

Step 2: Network Address

10.0.5.0

Step 3: Broadcast Address

10.0.5.255

Step 4: Valid Host Range

First Host : 10.0.5.1

Last Host : 10.0.5.254

Final Answer

Item	Answer
Network Address	10.0.5.0
Broadcast Address	10.0.5.255
First Host	10.0.5.1
Last Host	10.0.5.254
CIDR	/24

5. Use an online subnet calculator tool (such as <https://www.subnet-calculator.com/>) to verify your answer for the previous task. Take a screenshot of your calculation and write one line explaining if your manual answer matched the tool's output.

Enter

- **IP Address:** 10.0.5.17
- **Subnet Mask:** 255.255.255.0

Result

Network Address : 10.0.5.0

Broadcast Address : 10.0.5.255

Host Range : 10.0.5.1 – 10.0.5.254